Unit: Unit 3: Functions
Topic: Operations with Functions

Name: ______________
Date: __________

Advanced (Habanero)

Q1. Given $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?
(A) $x^2 + 2x$
(B) $x^2 - 2x$
(C) $2x^2$
(D) $x^2 + x$
(E) $2x^2 + x$

Q2. If $f(x) = x - 2$ and $g(x) = x + 2$, what is $(f - g)(x)$?
(A) -4
(B) $2x - 4$
(C) 4
(D) $2x + 4$
(E) 0

Q3. Given $f(x) = x^2$ and $g(x) = 2x$, what is $(f \cdot g)(x)$?
(A) $2x^3$
(B) $2x^2$
(C) $x^2 + 2x$
(D) $2x^2 + x$
(E) x^3

Q4. If $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x+1}$, what is $(f \cdot g)(x)$?
(A) $\frac{1}{x(x+1)}$
(B) $\frac{x+1}{x}$
(C) $\frac{x}{x+1}$
(D) $x(x + 1)$
(E) $\frac{1}{x^2}$

Q5. Given $f(x) = x^2$ and $g(x) = x + 2$, what is the domain of $(f + g)(x)$?
(A) $(-\infty, \infty)$
(B) $(-\infty, -2)$
(C) $(-2, \infty)$
(D) $(-2, 2)$
(E) $[0, \infty)$

Q6. If $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$, what is the domain of $(f + g)(x)$?
(A) $(-\infty, \infty)$
(B) $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$
(C) $(-\infty, 0) \cup (0, \infty)$
(D) $(-\infty, 1) \cup (1, \infty)$
(E) $[0, 1)$

Q7. Given $f(x) = x^2 - 1$ and $g(x) = x^2 + 1$, what is $(f - g)(x)$?
(A) -2
(B) $2x$
(C) $2x^2$
(D) $2x^2 - 2$
(E) $-2x^2$

Q8. If $f(x) = x^2$ and $g(x) = 2$, what is $(f \cdot g)(x)$?
(A) $2x$
(B) $2x^2$
(C) $x^2 + 2$
(D) $2x^2 + x$
(E) x^2

Open Ended Questions (FRQ)

Q9. Simplify $(f + g)(x)$ given $f(x) = x^2 - 2x + 1$ and $g(x) = x^2 + x - 1$

Q10. Determine the domain of $(f \cdot g)(x)$ given $f(x) = \frac{1}{x-2}$ and $g(x) = \frac{1}{x+2}$

Chef's Tips

- Make sure students show all work and simplify their answers.
- Remind students to check their answers for any domain restrictions.
- Emphasize that students must use the correct operations when combining functions.